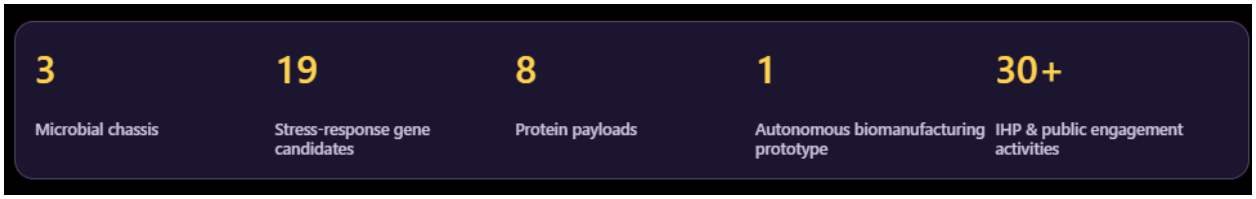
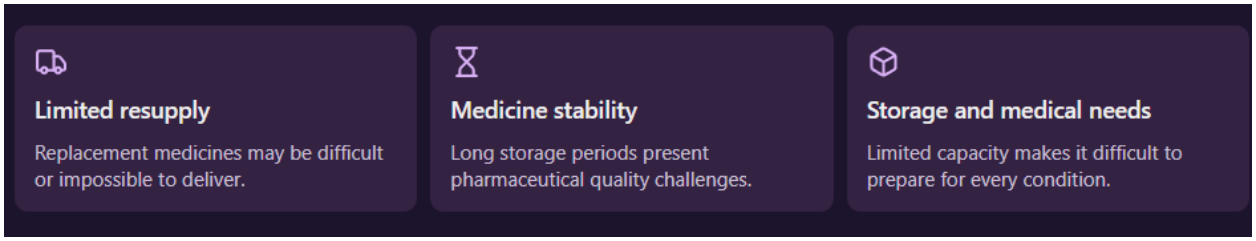
SHIHAB (Synthetic Hosts for In-situ Healthcare and Autonomous Biomanufacturing) explores a new approach to space healthcare by combining resilient microbial hosts with a compact, autonomous platform for biological production and purification. Our goal is to enable the on-demand production of selected biologics during long-duration space missions, supporting greater medical autonomy beyond Earth.



The problem: Medicine beyond Earth



As humanity prepares for longer and more ambitious space missions, ensuring that astronauts' health and well-being are supported throughout these journeys becomes even more important. Missions such as NASA's Expedition 75 and future Lunar exploration initiatives highlight the growing role of astronauts in conducting scientific research beyond Earth [1,2]. Today, astronauts aboard the International Space Station (ISS) have access to medical supplies containing medications for various anticipated health conditions. These can be replenished through cargo missions, and evacuation to Earth in case of a serious medical emergency remains possible. However, greater distances and longer mission durations will make these support systems difficult to rely on as exploration extends toward the Moon, Mars, and beyond [3,4]. astronaut-medicine



Limited resupply	**Medicine stability**	**Storage and medical needs**
On the ISS, medical supplies	Although medications stored	Every kilogram and cubic

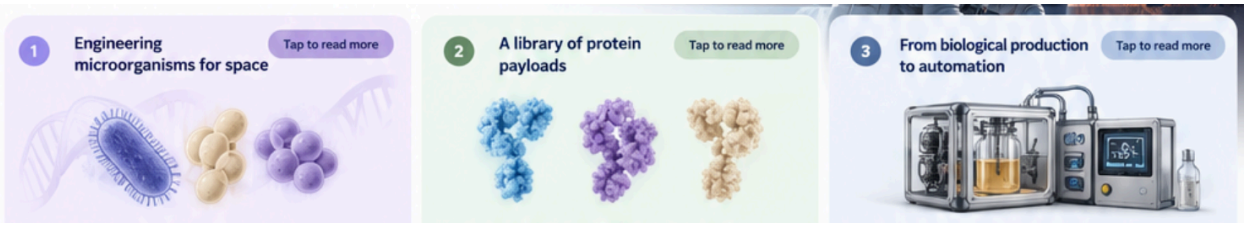
can be replaced through resupply missions. For exploration missions farther from Earth, however, replacement medications may take months to arrive or may not be available at all. Astronauts will need to rely more heavily on the resources they carry at launch, making medical planning and onboard autonomy increasingly important [3].	aboard the ISS have generally remained usable, some studies have identified stability concerns for particular pharmaceuticals. As missions extend over several years, prolonged storage combined with spaceflight-specific environmental conditions (ionizing radiation, altered gravity, and temperature fluctuations) may increase the difficulty of ensuring consistent pharmaceutical quality and effectiveness [4,5].	centimeter aboard a spacecraft matters. Medical supplies must compete with other essential resources for limited mass and storage capacity. At the same time, it is impossible to predict every illness, injury, or medical complication astronauts may encounter. Carrying sufficient quantities of every potentially necessary medication would be impractical, requiring careful decisions about which treatments to prioritize [3,6].

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Our approach: In-situ biomanufacturing

We developed SHIHAB to explore a new approach to space healthcare, where astronauts could produce selected biologics on demand rather than relying completely on conventional pharmaceutical supplies. In this regard, we aim to give astronauts greater independence from Earth-based infrastructure during long-duration missions.

Click on the tree cards below to explore SHIHAB's key components.



Card 1 organisms
Engineering microorganisms for space (Tap to read more)
Our approach focuses on engineering microbial hosts capable of functioning under space-associated environmental stressors, including ionizing radiation, oxidative stress, and temperature fluctuations. We selected *Bacillus subtilis*, *Saccharomyces cerevisiae*, and *Pichia pastoris* as promising chassis due to their established applications in biotechnology. To improve their resilience, we genomically integrated Dsup, a DNA-associated protein found in tardigrades that has demonstrated protective effects against radiation-induced DNA damage and oxidative stress tardigrades. We also propose preserving the microorganisms in compact formats,

such as lyophilized preparations or spores where biologically appropriate, until needed for protein production.

Additionally, inspired by the extreme conditions of Saudi Arabia's Rub' al Khali desert, our dry-lab team developed a gene-mining pipeline that identified 19 candidate stress-response genes associated with DNA repair, cold-shock adaptation, and desiccation tolerance.
dessert.jpg

We aimed to develop robust and resilient microbial hosts capable of supporting reliable biomanufacturing under challenging environmental conditions. (Go to the next card)
Button to our Wet Lab Design

Card 2 proteins
A library of protein payloads (Tap to read more)
Astronauts rely on medications transported from Earth, which may become increasingly difficult to replenish during farther missions.
Our engineered microorganisms are intended to function as biological factories capable of producing therapeutic proteins and other valuable biomolecules for space applications. To explore this potential, we developed a library of protein payloads with applications in astronaut health, scientific research, and biotechnology.

Our library includes therapeutic nanobodies, such as anti-PD-L1 and Nb1_gbHSV, targeting immune checkpoint pathways and herpes simplex virus, respectively; antifreeze proteins from the Antarctic bacterium *Flavobacterium frigoris* and Arctic yeast *Leucosporidium sp. AY30*, with potential applications in cryopreservation; and fluorescent reporters, including GFP and mCherry, to evaluate our expression systems. The system is not limited to these payloads, future works will include expressing other therapeutic candidates, such as proteins related to treatment of muscle atrophy and bone loss. (Go to the next card)

Button to our Protein Library

Card 3 suitcase
From biological production to automation (Tap to read more)
Efficient expression and purification of proteins require much more than engineering of the chassis. Microorganisms must grow under controlled favourable conditions, payload production must be monitored and induced, and the resulting molecules must be recovered and purified. Performing these steps manually would place additional demands on astronauts, whose time and resources are already limited. Moreover, it is highly costly and challenging to establish a whole laboratory with centrifuges, purification machines, and other equipment in a spacecraft. To address this, we designed a compact, **autonomous biomanufacturing platform** that integrates a bioreactor and automated downstream purification system. Designed with spaceflight constraints in mind, the system aims to minimize crew intervention through sensor-based monitoring, automated control of cultivation conditions, induction of protein production, and purification of the resulting biologics.

Button to our Hardware

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Beyond the lab: Shaping SHIHAB together

(Design) Can you make a slider of photos?

meetingall.jpeg
sarah-meeting.jpeg
rayannah-meet.jpeg
luis-meeting.jpeg
andrew-meeting.jpeg
PE-activity.jpeg
impact.jpeg
pe-activity

Developing SHIHAB required us to look beyond the laboratory and understand the people, environments, and communities our technology could eventually serve. Through our Integrated Human Practices, we engaged with astronauts, space medicine professionals, scientists, industry representatives, and regulatory experts, whose perspectives helped us reconsider our priorities, identify limitations, and guide decisions throughout the project's development.

But responsible innovation also requires involving society in discussions about emerging biotechnologies. Through the Tawasol Hackathon, 186 students explored biosafety, public trust, ethics, and accountability. The Tawasol Summit brought together iGEM teams from four countries to discuss regional perceptions of synthetic biology, while our collaboration with Kallpa Greenz in Peru explored community concerns surrounding genetic engineering and supported educational initiatives for rural communities. Alongside these initiatives, our Space Academy and mentorship activities aimed to make space science and biotechnology more accessible, encouraging the next generation to explore, question, and contribute to scientific innovation.

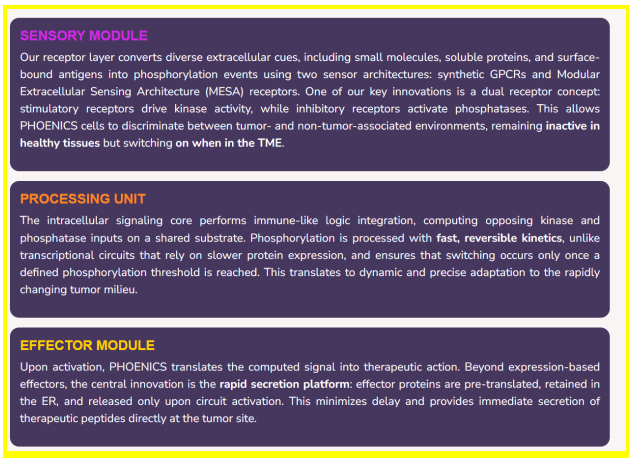
Our efforts gave us an important lesson, responsible innovation requires more than technological progress. It also relies on education, open dialogue, and the active participation of the communities that emerging technologies may one day impact.

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Our vision

What started as an exploration of microbial resilience evolved into a broader vision of making biological production possible in environments where conventional pharmaceutical supply chains are difficult to maintain.

Design



Beyond Earth

As human exploration extends toward the Moon, Mars, and beyond, the ability to maintain crew health with fewer resources from Earth will become increasingly important. We envision a future in which astronauts could access a library of preserved microbial production strains, selecting the biological payload they need and using a shared manufacturing platform to produce it during their mission. Such a system could complement conventional medical supplies, offering greater flexibility in responding to healthcare needs while reducing dependence on transporting every therapeutic product in its finished form.

Back on Earth

Although SHIHAB was conceived for space exploration, the challenges it addresses are not exclusive to space. Remote communities, disaster-affected regions, and locations with limited pharmaceutical infrastructure can also experience difficulties accessing essential medicines.

The principles behind compact, modular, and automated biomanufacturing could eventually support decentralized production systems capable of manufacturing selected biologics closer to where they are needed. By exploring these possibilities, SHIHAB connects the challenges of space exploration with opportunities to improve access to biotechnology on Earth.

Looking ahead

Realizing this vision will require further advances in microbial preservation, reliable protein production, automated purification, product quality control, and system validation under relevant spaceflight conditions. SHIHAB represents an early step toward exploring how these capabilities could work together.

(End of design)

Our vision is a future in which access to biological therapeutics is no longer defined solely by distance from Earth or proximity to conventional pharmaceutical infrastructure.